

```
yuktha@LAPTOP-UAVR6HQ0:~$ gedit q3_history.c

** (gedit:465): WARNING **: 03:28:38.127: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:465): WARNING **: 03:28:38.127: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
yuktha@LAPTOP-UAVR6HQ0:~$ sudo apt update
[sudo: authenticate] Password:
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [440 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [565 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [104 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.6 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [169 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [52.9 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [361 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [70.0 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [134 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [98.3 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [263 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [86.3 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [188 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [371 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [71.7 kB]
Get:21 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [11.3 kB]
Get:22 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [3032 B]
Get:23 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Packages [3132 B]
Get:24 http://archive.ubuntu.com/ubuntu resolute-backports/universe Translation-en [7988 B]
Get:25 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1056 B]
Fetched 3501 kB in 50s (69.6 kB/s)
102 packages can be upgraded. Run 'apt list --upgradable' to see them.
yuktha@LAPTOP-UAVR6HQ0:~$ sudo apt install libreadline-dev
Installing:
  libreadline-dev

Installing dependencies:
  libncurses-dev

Suggested packages:
  ncurses-doc readline-doc

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 102
  Download size: 570 kB
```

```

Continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libncurses-dev amd64 6.6+20251231-1 [389 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libreadline-dev amd64 8.3-4 [180 kB]
Fetched 570 kB in 18s (32.0 kB/s)
Selecting previously unselected package libncurses-dev:amd64.
(Reading database ... 48243 files and directories currently installed.)
Preparing to unpack .../libncurses-dev_6.6+20251231-1_amd64.deb ...
Unpacking libncurses-dev:amd64 (6.6+20251231-1) ...
Selecting previously unselected package libreadline-dev:amd64.
Preparing to unpack .../libreadline-dev_8.3-4_amd64.deb ...
Unpacking libreadline-dev:amd64 (8.3-4) ...
Setting up libncurses-dev:amd64 (6.6+20251231-1) ...
Setting up libreadline-dev:amd64 (8.3-4) ...
Processing triggers for install-info (7.2-Subuntu2) ...
Processing triggers for man-db (2.13.1-1build1) ...
yuktha@LAPTOP-UAVR6HQ0:~$ gcc q3_history.c -o q3_history -lreadline
yuktha@LAPTOP-UAVR6HQ0:~$ ./q3_history
=== Mini Shell History Test ===
Type commands. Type 'exit' to quit.

myshell> hello
Command entered: hello
myshell> ls
Command entered: ls
myshell> pwd
Command entered: pwd
myshell> pwd
Command entered: pwd
myshell> exit

Command History:
1: hello
2: ls
3: pwd
4: pwd
5: exit
yuktha@LAPTOP-UAVR6HQ0:~$ gedit q3_memory.c

** (gedit:1252): WARNING **: 03:33:25.310: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:1252): WARNING **: 03:33:25.310: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
yuktha@LAPTOP-UAVR6HQ0:~$ gcc -Wall -Wextra -g q3_memory.c -o q3_memory
yuktha@LAPTOP-UAVR6HQ0:~$ ./q3_memory
Enter initial array size: 3
Enter 3 elements:
10

```

```
20
30
Original array:
10 20 30
Enter new array size: 5
Enter 2 additional elements:
20
80
Resized array:
10 20 30 20 80

Creating linked list...
Enter value for node 1: 100
Enter value for node 2: 200
Enter value for node 3: 300

Linked List:
300 -> 200 -> 100 -> NULL

Memory released successfully.
```